

Mathematics

Decision-Making Support

IDENTIFICATION

CODE : IF-4-AD
ECTS : 1.0

HOURS

Lectures : 9.0 h
Seminars : 0.0 h
Laboratory : 8.0 h
Project : 0.0 h
Teacher-student
contact : 17.0 h
Personal work : 8.0 h
Total : 25.0 h

ASSESSMENT METHOD

written evaluation of the project
Final examen (1h30), documents on
paper can be used

TEACHING AIDS

Lecture slides
case studies and exercises
scientific papers

TEACHING LANGUAGE

French

CONTACT

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AIMS

The lecture objective is to develop skills aiming to:

- Analyse decision making problems, introducing basing concepts such as decision making factors, preferences, criteria, constraints.
- Select the most suitable mathematical representation according to an optimization objective and implement the solving, methods introducing basing concepts such as mathematical programming, optimization.
- Present and justify a mathematical approach to non-specialists with argumentation, synthesis

CONTENT

The course syllabus will address the 3 following main topics:

- Introduction and positioning of a decision making process
- Mathematical programming: problem main categories, non-constrained/constrained multidimensional optimization (linear/nonlinear), Simplex algorithm
- Multi-criteria decision making: multi-criteria nonlinear programming, discrete multi-criteria analysis, the ELECTRE method

BIBLIOGRAPHY

- [1] Minoux Michel. Programmation mathématique - Théorie et algorithmes Tomes 1 et 2, Dunod
[2] Roy Bernard. Méthodologie multicritère d'aide à la décision, Gestion Economica
[3] Scharlig Alain. Décider sur plusieurs critères, Collection Diriger l'entreprise, Presse Polytechniques et Universitaires Romandes